
Morby Ver1.0 — POC Deployment Summary

GenAI Mortgage Assistant + eKTP Screening (PARIKSA) + Lead CMS + Sales Portal

Morby Ver1.0 · the Bank Mortgage Buddy · Live POC on Cloudflare · Internal

1. Project Name

Morby Ver1.0 (the Bank Mortgage Buddy) — a browser-based GenAI mortgage (KPR) assistant with integrated eKTP/NIK screening (PARIKSA), a full lead-management CMS (mini-Salesforce), a per-region Sales portal, and customer self-service status tracking. Deployed end-to-end on Cloudflare (Workers + R2 + D1) with Google Gemini as the AI engine and Resend for email. Production target: on-premise servers with self-hosted AI.

2. Problem, Solution, and Stakeholders

Problem

- Repetitive KPR product/rate/promo questions consume relationship-manager time; answers vary in accuracy.
- Lead intake is manual and unstructured; eKTP identity capture is error-prone; leads are lost or followed up late.
- Sales follow-up has no SLA enforcement, no single pipeline view; management lacks real-time visibility; customers can't see their own status.
- Handling eKTP/NIK (specific personal data under UU PDP 27/2022) by email/manual copies creates compliance exposure.

Solution

- 24/7 Bahasa Indonesia (and English) chatbot grounded on the Bank knowledge base; a standalone installment/cashback calculator with a downloadable PDF (simulation, non-binding); document-checklist and promo PDFs served in chat.
- Guided prescreen (incl. requested plafon + tenor) with explicit recorded PDPA consent; AI Vision eKTP reading verified by a deterministic NIK validator. Privacy-by-design: the full eKTP scan is NOT stored — only extracted text + the cropped pas foto.
- Automatic CMS: dedup, qualification flags, 3-dimension scoring + grade, auto sales assignment by region (AS/HB/RB/ER), SLA tasks (call 30 min, WA 1 h) with cron reminders and a Friday weekly sweep.
- Role-based portals: admin Super Page (pipeline, documents, audit log, Customer-360 export, BI dashboard) and a scoped Sales portal (/sales) where each salesperson sees only their own leads and updates status with free-text notes.
- Customer self-service: a reference number returned at submission lets the applicant check their live application status in chat.

Stakeholders

Stakeholder	Role / Benefit
Prospective customers	Instant accurate answers; simple guided application; transparent consent; self-service status check
Sales (AS/HB/RB/ER)	Auto-assigned scored leads; SLA reminders; own pipeline portal with status + notes
Management / business	BI dashboard (volume Rp, take-up & approval rate); Customer-360 & monthly recaps
Admin / operations	Super Page: lead logs, document management, audit trail, password/reset controls
Compliance / DPO	Recorded consent, masked NIK in DB, no full eKTP scan stored, audit log, PDP/OJK audit docs

3. Business Value / Expected Outcome

Business impact

- Faster lead response: SLA tasks + automatic reminders replace ad-hoc follow-up; weekly sweep prevents stale leads.
- Higher-quality pipeline: every lead is deduplicated, scored (A+–D), and routed to the right region automatically.

Customer / user impact

- 24/7 self-service answers grounded on official terms; first token streams in ~1 second (AI 'thinking' disabled).
- One-visit application start (chat → prescreen → eKTP → submitted) typically under 10 minutes; customers can track status by reference number.

Revenue gain

- More leads captured (24/7 + lower drop-off) and faster follow-up on high-grade leads are expected to lift take-up rate — measured directly on the BI dashboard (disbursed / leads).

Cost reduction / efficiency

- Deflects repetitive Q&A from RMs/call center; prescreen and identity capture are fully automated (no manual re-typing).
- POC infrastructure cost near-zero (Cloudflare + Gemini free tiers). Automated recaps (XLSX, Customer-360) remove manual reporting.

4. Success Metrics (POC) and Measurement

#	Metric	Target	How measured
1	Chatbot answer accuracy	≥ 90% correct	Test-set review; all rates from knowledge_base.json (single source)
2	eKTP NIK extraction	≥ 95% exact 16-digit	PAKIKSA tool + deterministic NIK validator verdict per lead

#	Metric	Target	How measured
3	Lead completion time	< 10 min chat → submit	Session start/end captured per lead (recap XLSX)
4	SLA compliance	100% reminded ≤ 5 min of breach	Cron every 5 min; reminder timestamps + audit log
5	Take-up rate visibility	Real-time, matches D1	BI: take-up = disbursed / total leads; validated vs D1
6	Customer status self-service	Status by ref works	/api/status returns first name + status only
7	Data compliance posture	No full eKTP scan stored; consent 100% recorded	R2 = ektp_data.txt + pas foto only; consent timestamp in recap

5. POC Scope — In-Scope Features

Customer-facing

- Bilingual chatbot (RAG, streamed, fast); standalone calculator page with PDF export (simulation/non-binding); promo & document-checklist PDFs in chat; application-status check by reference number; pinned-composer chat UX.
- Explicit un-ticked PDPA consent + privacy PDF; eKTP AI-OCR (Gemini Vision, on-device fallback) → text fields + pas foto + NIK validator.

CMS, Sales & Operations (Phases 1–8)

- Ingest → D1+R2 (email fallback); dedup + qualification flags; scoring & sales assignment; SLA tasks + cron + weekly sweep; 10-status pipeline with history & free-text notes.
- Customer-360 multi-tab XLSX + pas-foto export; BI dashboard (sessions, YTD leads, customers, total limit, in-analyst, approval rate, disbursed, take-up rate; monthly Rp-volume / customer chart).
- Auth (first-login password hashed in D1; email temp-password reset), per-document management, tamper-logged audit trail; Sales portal (/sales) scoped per owner AS/ER/HB/RB.

Out of scope (POC)

- Core-banking / credit-engine integration; real credit decisioning; MFA/SSO; automated data-retention; on-prem deployment (designed in ON_PREM_ARCHITECTURE.md, not yet built).

6. Data Input & Output

Inputs

Input	Description	Source
Knowledge base	Products, rate table, promos 1064/1065/1066, fees, FAQ, eligibility	Bank product docs (curated)
Prescreen answers	Name, phone, email, income, occupation, requested plafon & tenor, city, credit history	Customer via guided chat
eKTP image (transient)	Read by AI Vision in-flight; full scan NOT stored	Customer upload + consent

Input	Description	Source
Status updates + keterangan	Pipeline changes with free-text notes	Sales / admin portals

Stored data & outputs

Output	Description
ektp_data.txt	Extracted eKTP text: nama, NIK, TTL, jenis kelamin, status perkawinan, provinsi, kab/kota, tanggal pembuatan
pasfoto.jpg	Auto-cropped face photo; full card scan never stored
Lead record (D1)	Ref, masked NIK, contact, product, plafon/tenor, grades + score, sales owner, SLA timestamps, status
Customer-360 / recap XLSX	4-tab 360 (Total + PRI/SEC/TO) and monthly recap with consent time & email status
BI dashboard + audit log	Live metrics (scoped per sales); every download/delete/status/auth event logged
Calculator PDF	Installment + cashback simulation with a non-binding disclaimer

7. Innovation, Differentiation & Tools

Innovation

- Hybrid AI + deterministic: GenAI handles language & vision, but every number (rates, installments, cashback, NIK validity) comes from deterministic code + one knowledge base — no hallucinated figures.
- Privacy-by-design identity capture: eKTP stored as text + pas foto only; full card scan never persists; consent recorded per lead.
- Chat-to-CRM automation: a conversation directly produces a scored, deduplicated, sales-assigned lead with SLA timers — plus customer self-service status.
- Self-healing, no-code ops: D1 auto-migrates columns; email fallback protects leads; Gemini model auto-discovery + fast first token; email temp-password reset.

Differentiation

- Full funnel in one lightweight serverless stack (chat → screening → CMS → SLA → pipeline → BI → exports), zero build step.
- Role-scoped portals (admin vs per-region sales) with audit logging from day one — bank-governance-ready.
- Clear on-prem production path (Sahabat-AI/SEA-LION/Qwen2.5 chat, Qwen2.5-VL eKTP OCR, BGE-M3 RAG) that resolves PDP cross-border & OJK localization, with a costed cloud-vs-on-prem comparison.

Tools & technology

Layer	Tools
Frontend	Vanilla HTML/CSS/ES modules (no build), strict CSP, self-hosted fonts
Backend / hosting	Cloudflare Workers (API + assets), Workers Builds CI, Cron Triggers

Layer	Tools
Storage	R2 (documents), D1/SQLite (CMS: leads, pipeline, audit, auth, metrics)
AI	Gemini Flash (chat, streamed, thinking off) + Gemini Vision (eKTP OCR JSON) + on-device Tesseract fallback
Deterministic core	NIK validator, annuity/cashback calculators, scoring & assignment rules
Email / docs	Resend (ZIP lead package, reminders, reset); OOXML XLSX, jsPDF PDFs
Security	HMAC HttpOnly sessions, hashed first-login passwords (none in repo), role gating, rate limiting, audit log

Internal document. Compliance conclusions require sign-off by the Bank's DPO, Legal, and Compliance. Live POC: <https://ef6af686-sakhapr.hendrik-panthron.workers.dev>